

# Seema Sultana

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## EDUCATION

- **Visvesvaraya Technological University** Bengaluru, India  
*Bachelor of Engineering - Computer Science And Engineering; GPA: 8.25* sept 2023 - present
- **Presidency PU College** Bengaluru, India  
*Class 12th - State Board; Grade: 92.5* 2021
- **SVN English High School** Bengaluru, India  
*Class 10th - State Board; Grade: 91.04* 2019

## SKILLS SUMMARY

- **Languages:** Python, JAVA, SQL, Typescript, Javascript
- **Frontend:** React.js, Next.js, Tailwind CSS
- **Backend:** Node.js, FastAPI, Django, Flask, REST APIs
- **Database:** PostgreSQL, MySQL, MongoDB
- **Frameworks:** NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, XGBoost, LangChain
- **Platforms:** Linux, Web, Windows, Arduino, AWS, IBM Cloud
- **Tools:** AWS (EC2, S3, Lambda, IAM, Bedrock, SageMaker), Git, GitHub, Docker
- **Distributed Systems:** Apache Kafka, Kubernetes, Nginx, Docker

## PROJECTS

- **Privacy-Preserving Federated Learning for IoT Intrusion Detection :** Project Link  
Engineered a PyTorch-Flower federated learning pipeline across 5 IoT clients, achieving 94.6% local accuracy under 2.0 Differential Privacy, integrating Opacus DP-SGD and Secure Aggregation to reduce training-data exposure.  
Tech: Python, PyTorch, Flower, Opacus, Scikit-learn, Pandas, NumPy, PyYAML, Joblib, Git/GitHub
- **FactSphere — Hallucination-Aware Multi-Agent Search Intelligence :** Project Link  
Accomplished 91% fabrication-detection accuracy across web platforms, improving search intelligence data trust scoring by 14% , through engineering a 4-agent AI architecture utilizing multi-search retrieval and evidence deduplication  
Tech: Python, Flask, Groq API, Llama 3.3 70B, BeautifulSoup, SentenceTransformers, ChromaDB, scikit-learn, Chrome Extension API (2026)
- **E-Commerce Distributed System Architecture:** Project Link  
Architected a distributed e-commerce platform supporting 25K+ concurrent users, reducing p95 latency by 30%+, by engineering event-driven microservices with Kafka, Redis caching, PostgreSQL replication/sharding, and Kubernetes autoscaling  
Tech: Python, Node.js, PostgreSQL, Redis, Docker, Kubernetes, Nginx, REST APIs, JWT, Apache Kafka

## EXPERIENCE

- **AWS Student builder group** On Campus  
*Co- lead (Full-time)* May 2026 - present
  - **Engineered cloud-native solutions:** using scalable AWS infrastructure tools.
  - **Delivered technical consultations:** to guide peers through cloud implementation strategies.
- **Embrizon Tecnologies - Data Science Internship** Remote  
*Data Science with AI - Certificate link* Feb - Apr 2026
  - **Consulted on technical delivery:** of end-to-end predictive analytics pipelines.
  - **Architected full-stack data dashboards:** to guide peers through cloud implementation strategies.

## HONORS AND AWARDS

- 1st Place – ELECTROLYTHON National Level Hackathon - Sept, 2025
- Top 6 – Gen-AI Hackathon - May, 2026
- Shortlisted – Google Build with AI (Bangalore) - Hackathon - June, 2026

## VOLUNTEER EXPERIENCE

- **AWS Student Builder Group - Vice President/Operations Lead** Bengaluru, India  
*Directed stakeholder communications between university leadership and developer communities.* July 2026 - Present
- **IEEE Computer Society Bangalore Chapter (IEEE CSBC)** Bengaluru, India  
*Managing the official IEEE CSBC website Coordinated IEEE CS PRO 2026, Girl Geek.* Jan 2026 - Present